

Interdisciplinary Entrepreneurship Portfolio Report

Integration of embedded, machine learning, path planning and vision technology | Pet intelligent products startup

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Learning Interest Process

October 2022

Get started with robots and try the Meglun embedded gripper robot 1st



February 2023

AMC Machine Learning Data Analysis



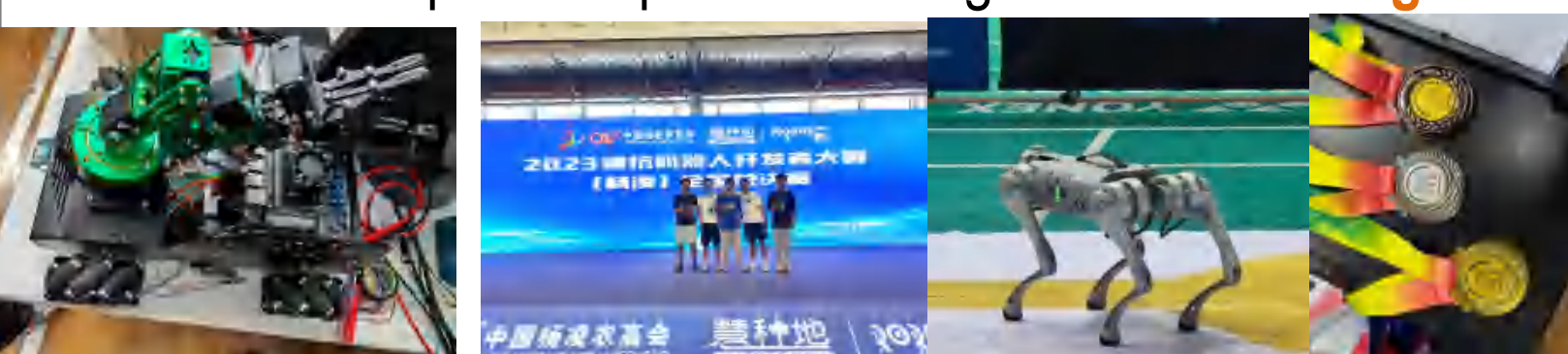
May 2023

Quadcopter Drone Challenge China Robot Skills Competition 1st



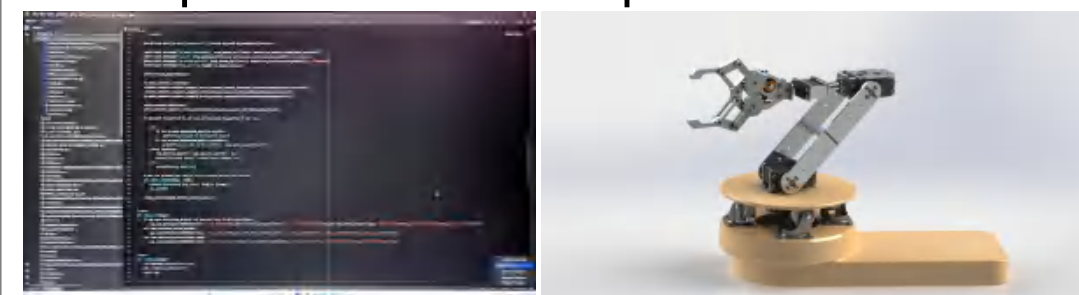
August 2023

RAICOM Developer Competition Intelligent Fruit Picking Robot 1st



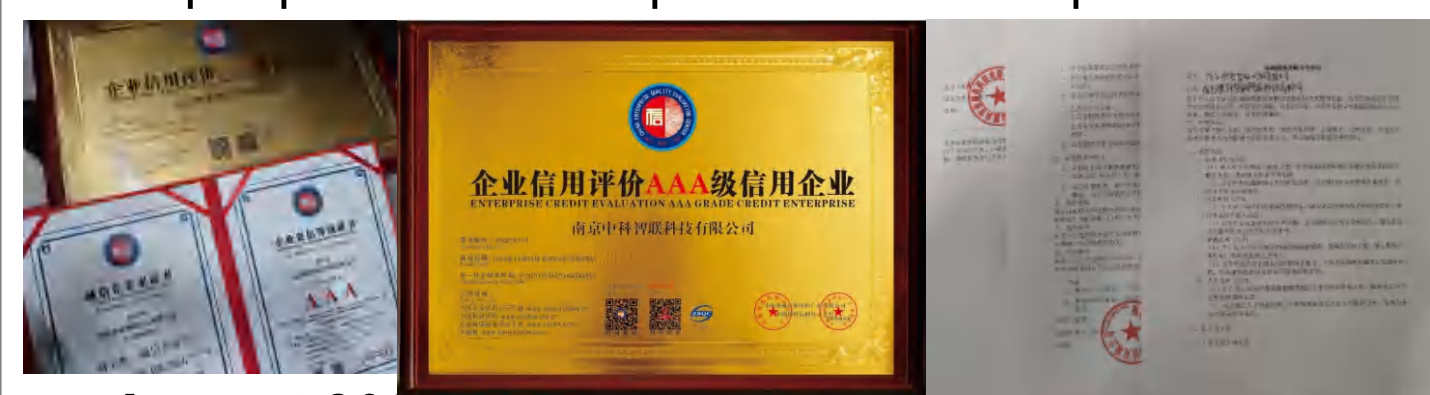
December 2023

Participate in other competitions and scientific research (eg. EI, IEEE, SCIE)



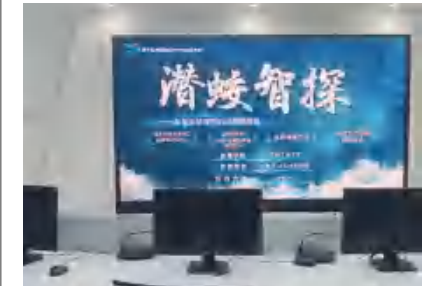
January 2024

Smart pet products startup start up business



August 2024

International College Student Innovation Competition Smart Sonar Submarine



September 2024 - May 2025

[1] Improved AI robot path planning published in **SCI** (online). [2] Machine learning in the interdisciplinary field of architecture is under review by SCI

May 2025 More deep learning and robotics research in progress (ECE)

Method 1

SCI paper result 1 JCR Q1 AI + Robot 1st author

An improved spider-bee optimizer for obstacle avoidance path planning of mobile robots <https://doi.org/10.3390/math12172604>

Breakthrough innovation	Zero collision in multiple scenarios	80% fault tolerance rate
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Interdisciplinary innovation: Integrating **bionics + machine learning** to solve the industry problem of "local optimum trap" in dynamic environments
Technical breakthrough: The "double midpoint guidance strategy", 100% avoiding obstacles (verified on 5 types of maps)

Self-developed population learning strategy - enabling robots to have "memory of failures"

Double midpoint hunting navigation - like a sniper calculating trajectory to predict obstacles Industry value

- The widespread application of mobile robots holds **significant importance for advancing social intelligence**. However, as the complexity of the environment increases, existing Obstacle Avoidance Path Planning (OAPP) methods tend to fall into local optimal paths, compromising reliability and practicality. Therefore, based on the Spider-Wasp Optimizer (SWO), this paper proposes an improved OAPP method called the LMBSWO to address these challenges. Firstly, the learning strategy is introduced to enhance the diversity of the algorithm population, thereby improving its global optimization performance. Secondly, the dual-median-point guidance strategy is incorporated to enhance the algorithm's exploitation capability and increase its path searchability. Lastly, a better guidance strategy is introduced to enhance the algorithm's ability to escape local optimal paths. Subsequently, the LMBSWO is employed for OAPP in five different map environments. The experimental results show that the LMBSWO achieves an advantage in collision-free path length, **with 100% probability**,

- Integration of embedded, machine learning, path planning and vision technology
- Pet intelligent products startup

- across five maps of different complexity, while obtaining 80% fault tolerance across different maps, compared to nine existing novel OAPP methods with efficient performance. The LMBSWO ranks first in the trade-off between planning time and path length. With these results, the LMBSWO can be considered as a robust OAPP method with efficient solving performance, along with high robustness.
- Keywords: spider-wasp optimizer; **learning strategy**; dual-median-point guidance strategy; better guidance strategy; **path planning**

The main contributions of this paper are as follows:

- The **application of eight novel algorithms** to solve the OAPP has introduced new perspectives and criteria for solving OAPP.
- The introduction of the learning strategy in SWO has enhanced the population diversity of the algorithm, thereby improving its capability to search for global optimal paths.
- The introduction of the dual-median-point guidance strategy in SWO has
- When animals were allowed to settle in plethysmographs for 15 enhanced the algorithm's path search capability.
- The introduction of the better guidance strategy in SWO has enhanced the algorithm's ability to escape from local optima in path search.
- The **LMBSWO-based OAPP** method is proposed in combination with the above strategies, achieving outstanding results in five different map environments.

Figure 1: LMBSWO model flow chart

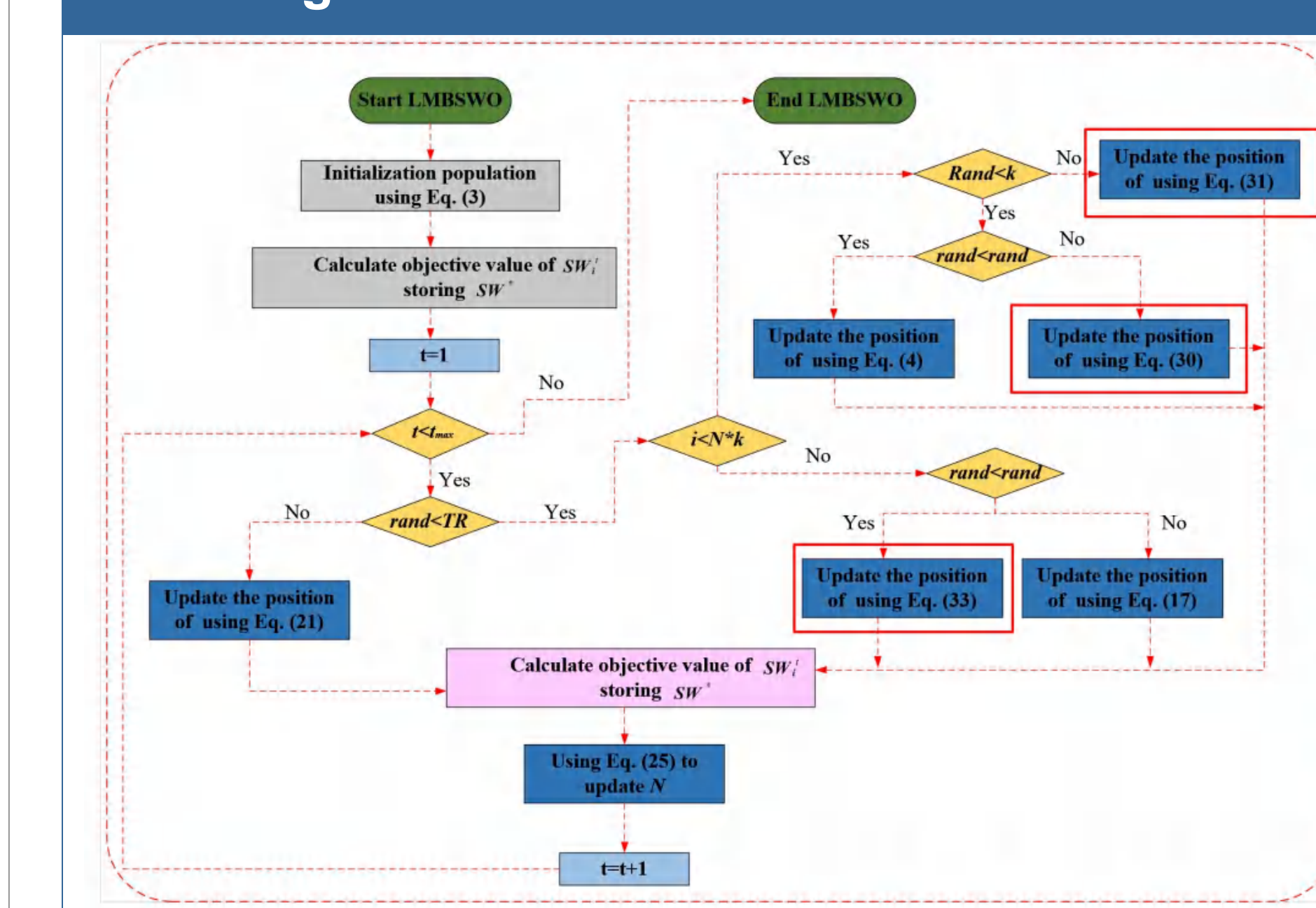


Figure 2: The dual-median-point guidance strategy simulation chart

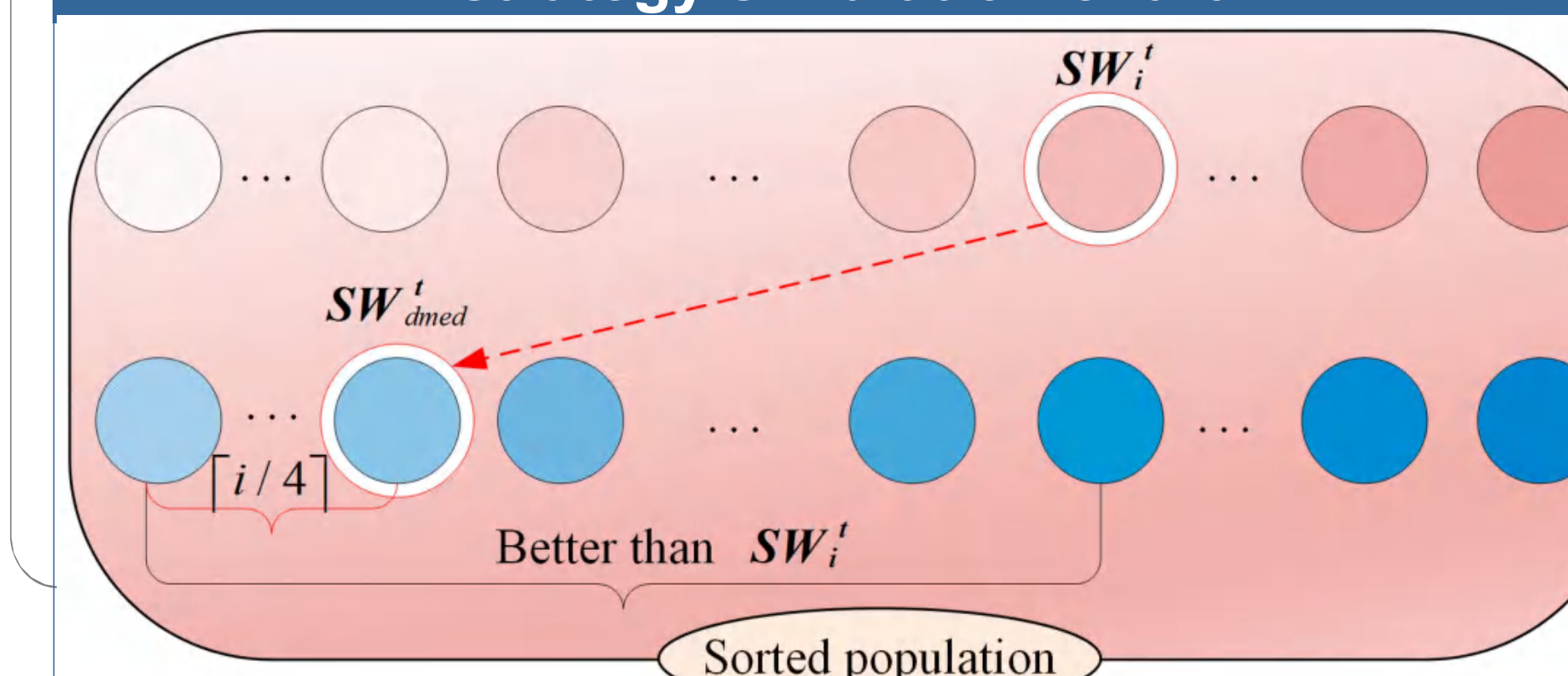


Figure 3: The better guidance strategy simulation chart

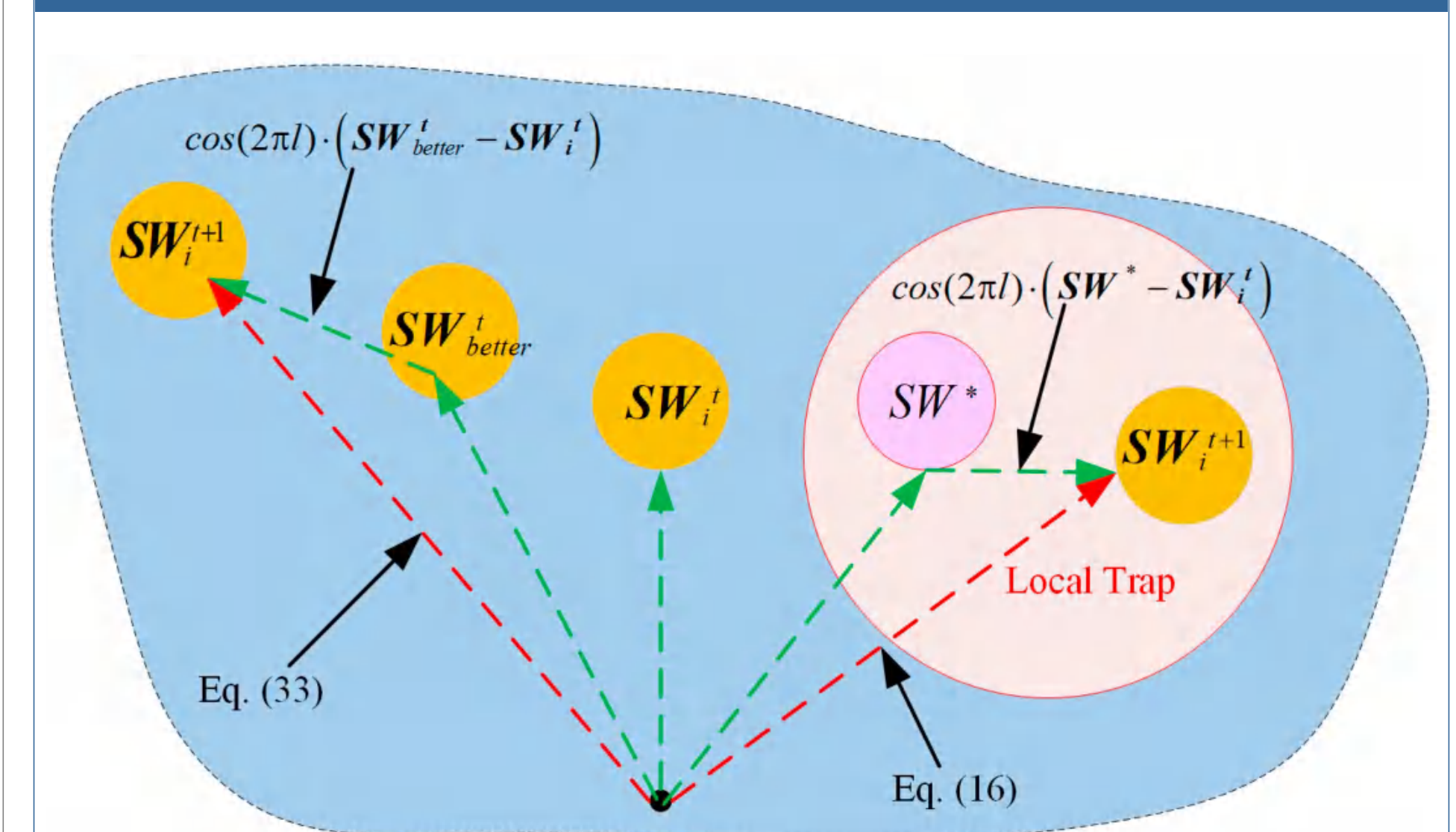
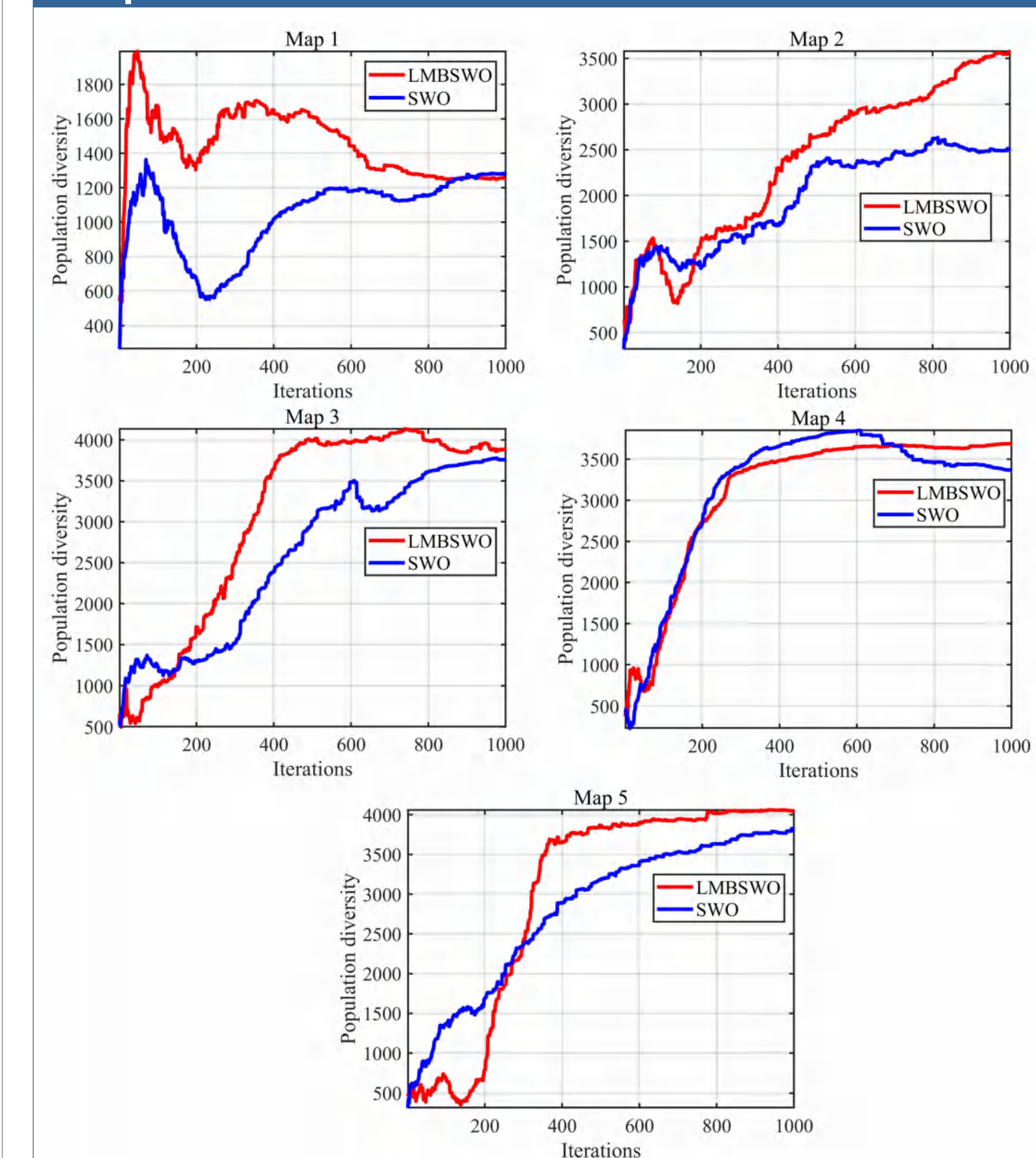


Figure 4: Population diversity across different maps



Algorithms	Times	Parameters
Artificial Hummingbird Algorithm (AHA) [29]	2022	Migration coefficient = $2 \cdot n$
Artificial Rabbits Optimization (ARO) [30]	2022	$L = (e^{(u-1)/T^2}) \cdot \sin(2\pi r)$, $H = (T - t + 1) / T \cdot r$
Dandelion Optimizer (DO) [31]	2022	$\alpha \in [0, 1]$, $k \in [0, 1]$
Exponential Distribution Optimizer (EDO) [32]	2023	switch parameter = 0.5
Weighted Mean of Vectors (INFO) [33]	2022	$c = 2$, $d = 4$
Kepler Optimization Algorithm (KOA) [34]	2023	$\mu_0 = 0.1$, $\gamma = 15$, $T = 3$
Northern Goshawk Optimization (NGO) [35]	2021	$R = 0.02 \cdot (1 - t/T)$
Particle Swarm Optimization (PSO) [36]	1995	$w = 1$, $w_p = 0.99$, $c_1 = 1.5$, $c_2 = 0.01$
Spider-Wasp Optimizer (SWO) [26]	2023	$CR = 0.2$, $TR = 0.3$
LMBSWO	NA	$CR = 0.2$, $TR = 0.3$

Table 1 Comparison algorithm parameter information(LMBSWO)

Every equation I write whispers a question: How can machines perceive the world with human-like agility?

Longitudinal research

Robot path planning and computer vision

Figure 5: The optimal path of LMBSWO across different maps

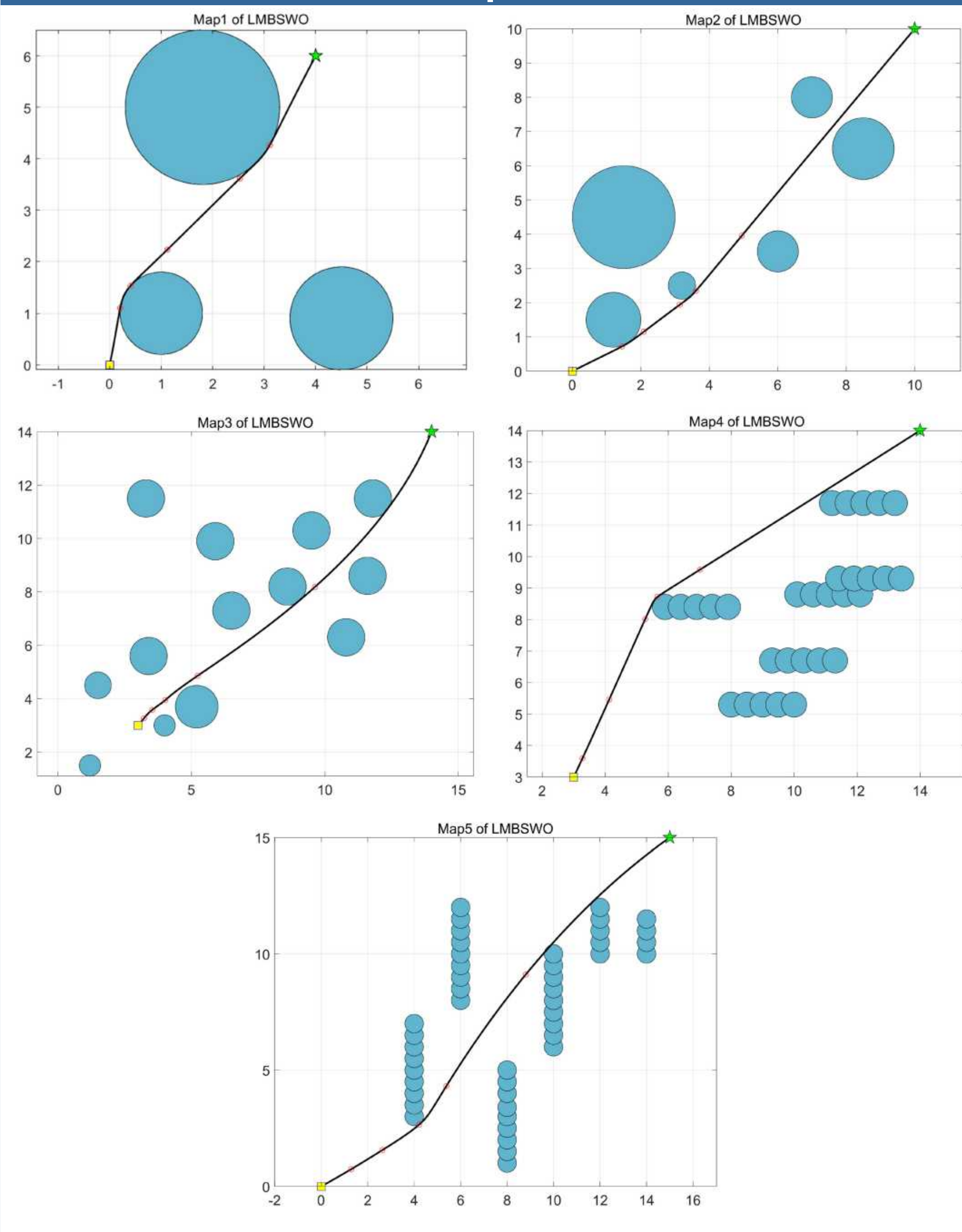
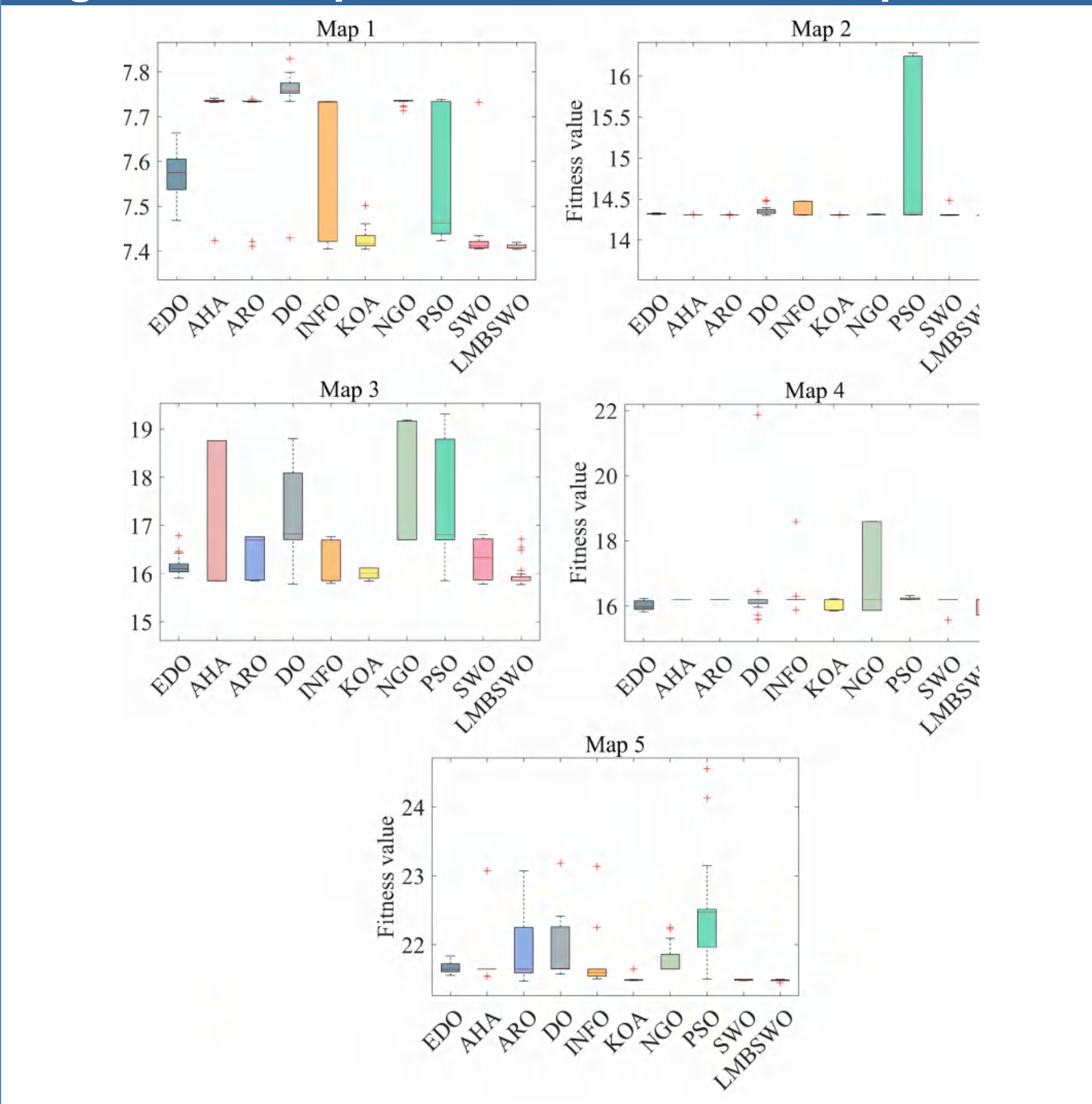


Figure 6: Box plots across different maps



This shows that the global optimization performance of the LMBSWO is **better compared to that of the state-of-the-art algorithms**.

Figure 7: Convergence plots across different maps

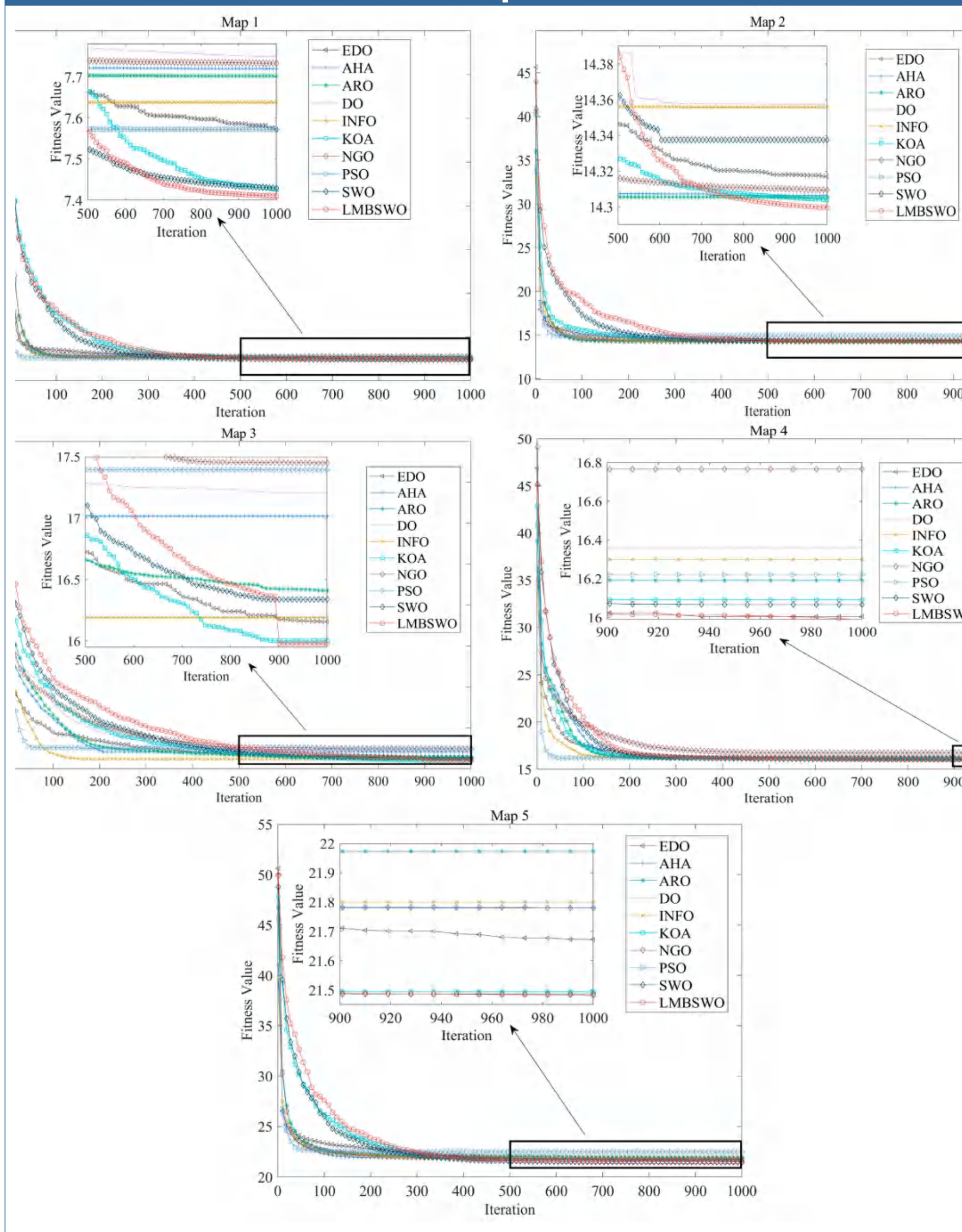
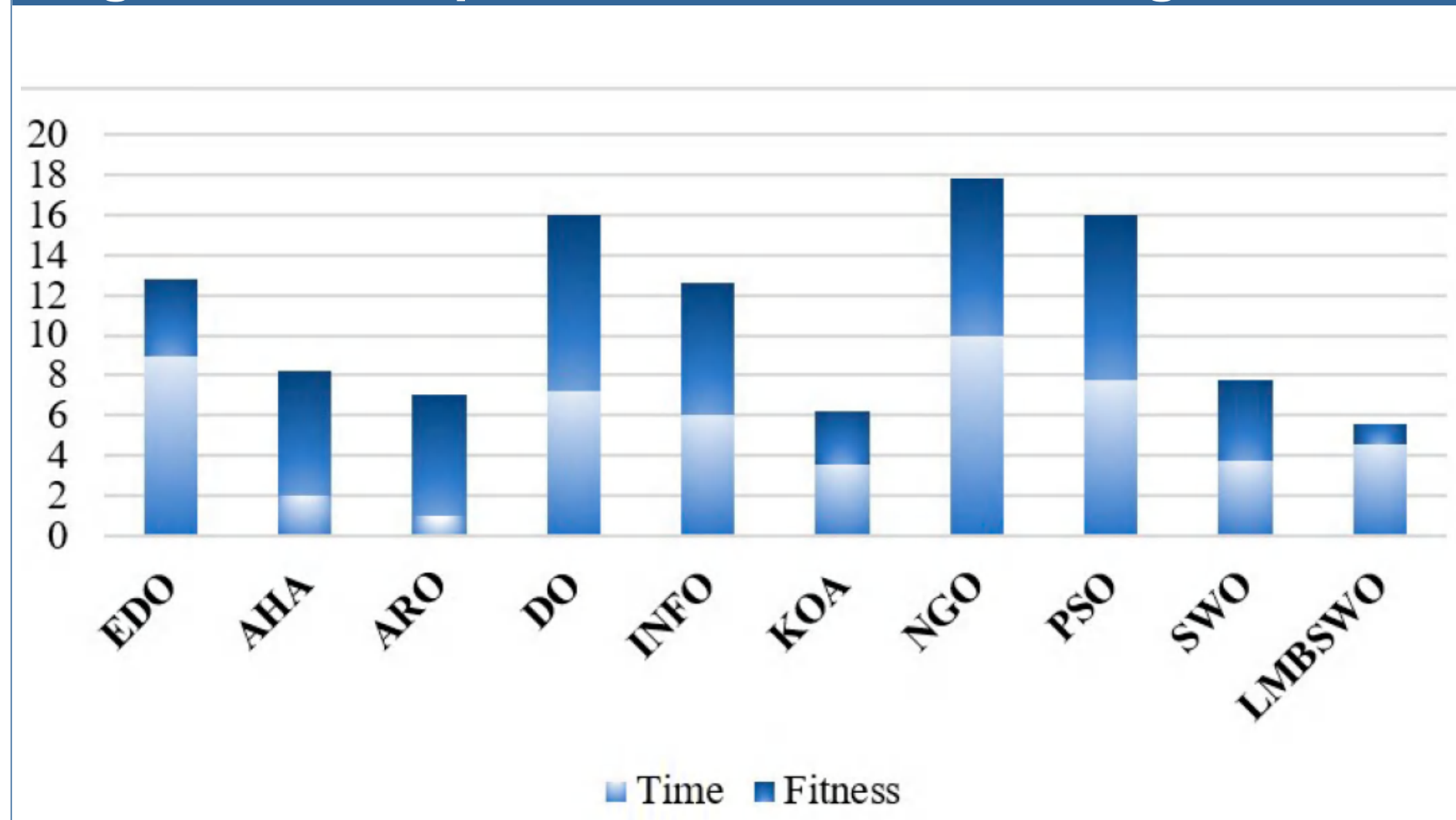


Figure 8: Composite indicator stacking chart



● **Enable robots to predict the blind zones of obstacles like humans do.**

● Figure 8 shows a stacked plot of the algorithm's average path length rankings and average runtime rankings when running path planning for the five map environments, where larger values on the Y-axis indicate poorer comprehensive algorithm performance.

Method 2

Research on **robot autonomous inspection visual perception system** based on improved **YOLO V11** and **SLAM**

AI + Vision

Oral presentation at an international conference

45 FPS real-time detection	Zero interference in dynamic environments	Energy consumption reduced by 45%
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Technology integration: Improving **YOLO V11 + Semantic SLAM, GhostNet**

lightweight , P - FPN, pioneering device state semantic mapping

Task adaptive mechanism: Reducing computational power consumption by 45% (through dynamic resolution switching)
Final results: The accuracy of small target recognition increased by 17.3%, and the consistency of SLAM mapping improved by 22.8%.

Aiming at the problems existing in the robot autonomous inspection system in complex scenarios, such as visual blind spots, dynamic environment interference, and insufficient algorithm robustness, this study proposes a robot visual perception system based on improved **YOLO V11** and **semantic SLAM**. Firstly, by introducing GhostNet lightweight convolution and the **Progressive Feature Pyramid Network (P - FPN)**, the YOLO V11 architecture is optimized. This achieves a 30% increase in detection speed (**45 FPS**) on the embedded platform (Jetson AGX) while maintaining an **mAP@0.5** accuracy of **92.1%**. Secondly, a semantic - enhanced SLAM framework is designed. Combining the dynamic region mask with the improved ORB - SLAM3 feature - matching strategy, the Absolute Trajectory Error (ATE) is reduced to 0.12m, and a hierarchical point - cloud map containing the semantic information of equipment status is constructed. In addition, a task - adaptive collaboration mechanism is proposed. According to the SLAM positioning information, the detection frequency and resolution mode are dynamically adjusted, reducing the system's computational resource **consumption by 45%**. Experiments show that on the self - built industrial inspection dataset, compared with the baseline combination (YOLOv11 + Cartographer), the small - target detection accuracy (AP - S) of this system is **improved by 17.3%**, and the SLAM mapping consistency score is **increased by 22.8%**. The actual deployment verifies the high - efficiency of the system in closed - loop tasks such as autonomous path planning, equipment status recognition, and abnormal alarm, providing an innovative solution for intelligent security in complex scenarios.

Figure 9: YOLOv11 architecture

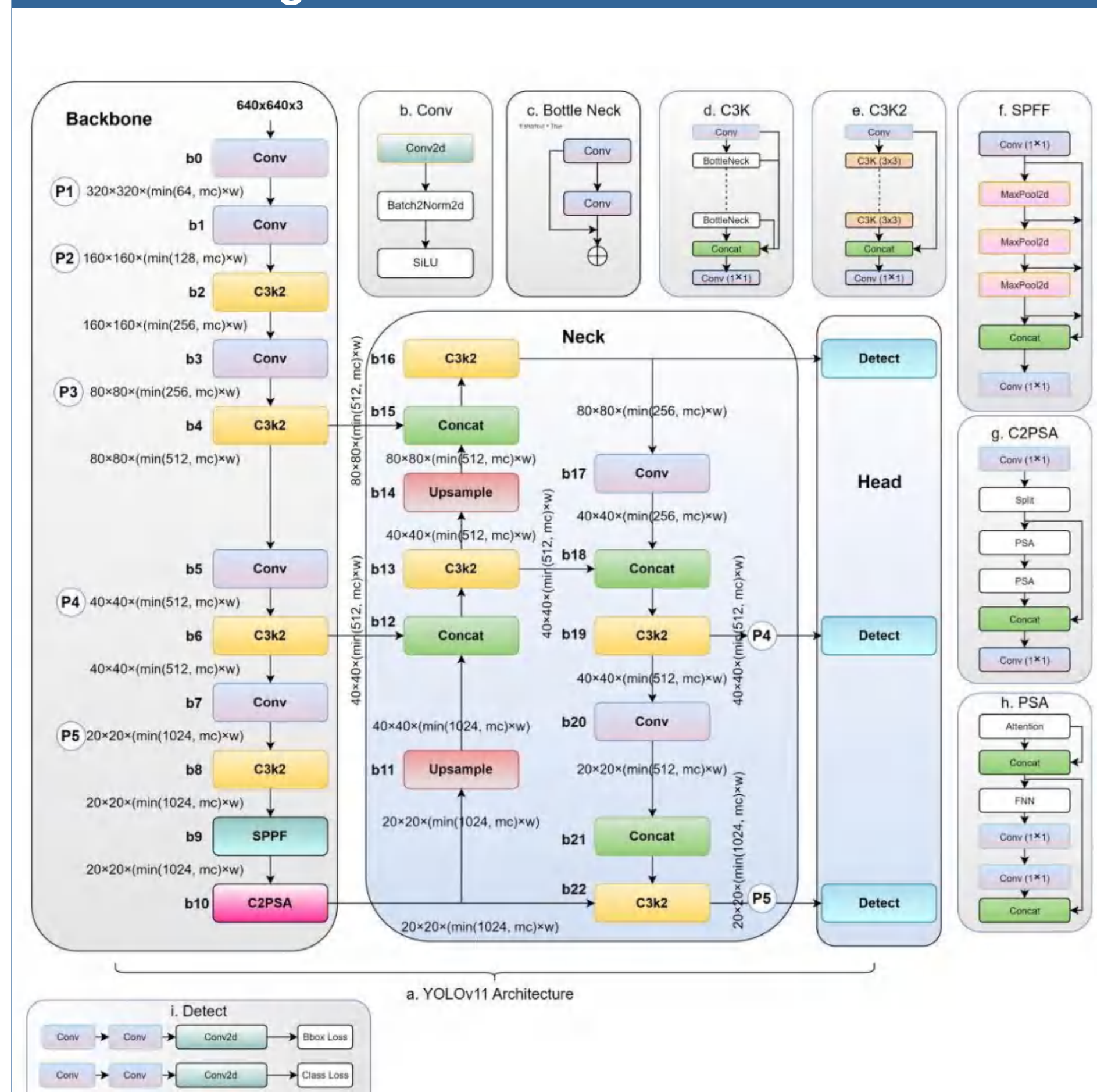
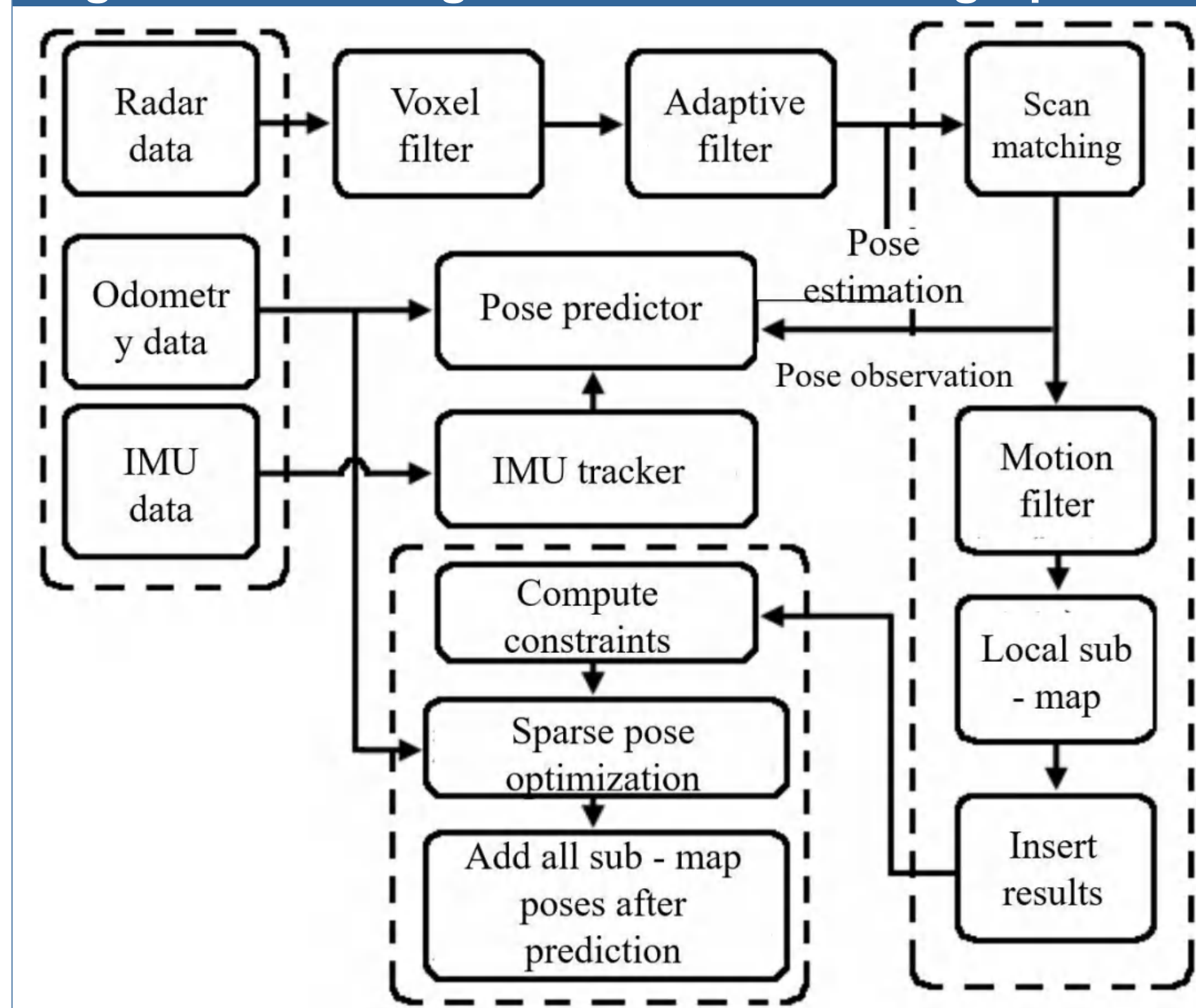


Figure 10: The Algorithm Flow of Cartographer



- When comparing the YOLOv11 improved based on GhostNet with the original YOLOv11, the accuracy of the improved version is slightly decreased, as shown in Figure 11.
- To test the integration effect of the improved **YOLOv11** and the **SLAM** algorithm, this experiment plans to place burning candles at a distance of n meters beside points A, B, and C respectively in the experimental scenario to simulate small - scale indoor flames. A route is set from point A to point B, and finally to point C. The test aims to verify whether the robot can accurately reach the navigation points and identify the flames, as shown in Figure 11.

Interdisciplinary Academic Research in robotics, AI, architecture, civil engineering, and computer vision



Table 2. Experiment on Positioning Accuracy

Error	Point Position	Error Value
Lateral Deviation/m	A	0.051
	B	0.054
	C	0.057
Longitudinal Deviation/m	A	0.037
	B	0.046
	C	0.076
Heading Deviation/(°)	A	8.023
	B	11.649
	C	15.056

Install an industrial doctor's eyes that can perform diagnostic checks on robots

Method 3

SCI paper result 2 JCR Q1
AI + Buildings 1st author
Behaviour prediction of connections in thin-walled steel-ply-bamboo structures based on machine learning for mechanical properties

Cross-border breakthrough	249 sets of connection components' full life cycle modeling	Selection efficiency increased by 5 times
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Turn structural member into a talking prophet
Discipline Integration Paradigm:
✓ Prediction of mechanical properties of thin-walled steel-cable composite structures (Civil Engineering × Machine Learning)
✓ Bayesian hyperparameter optimization: The classification accuracy of the RF model exceeds that of traditional experimental methods
Sustainable Value: Reducing Experimental Waste in Building Materials, Promoting Digitalization of Green Buildings
• This study employed multiple machine learning and hyperparameter optimization techniques to model and analyze the mechanical properties of self-drilling screw connections in thin-walled steel-ply-bamboo shear walls. The dataset which included 249 sets of measurement data, derived from 51 disparate connection specimens. Utilizing factor analysis, a ranking table recording comprehensive score of each connection specimens was established to help select the optimal connection type. Eight machine learning models were employed to analyse and predict the mechanical performance of these connection specimens. Through comparison, the most efficient model was selected and five hyperparameter optimization algorithms were implemented to further enhance its prediction accuracy. The analysis results obtained in this study revealed that the Random Forest (RF) was the optimal model with superior classification performance prediction accuracy and generalization ability. In hyperparameter optimization algorithms, the model processed through Bayesian Optimization (BO) outperforms both its pre-optimization state and models optimized using other algorithms. For the mechanical performance of connections within TWS composite structures, applying the BO algorithm to the RF model demonstrated a significant improvement in predictive accuracy. This approach enables the identification of the most suitable specimen type based on newly provided mechanical performance parameter sets.
• Industrialization integration implementation: Currently, this technology is being transformed into a patent for fire extinguishing technology in hospitals, in collaboration with renowned professor Zhang Yudong from Southeast University.

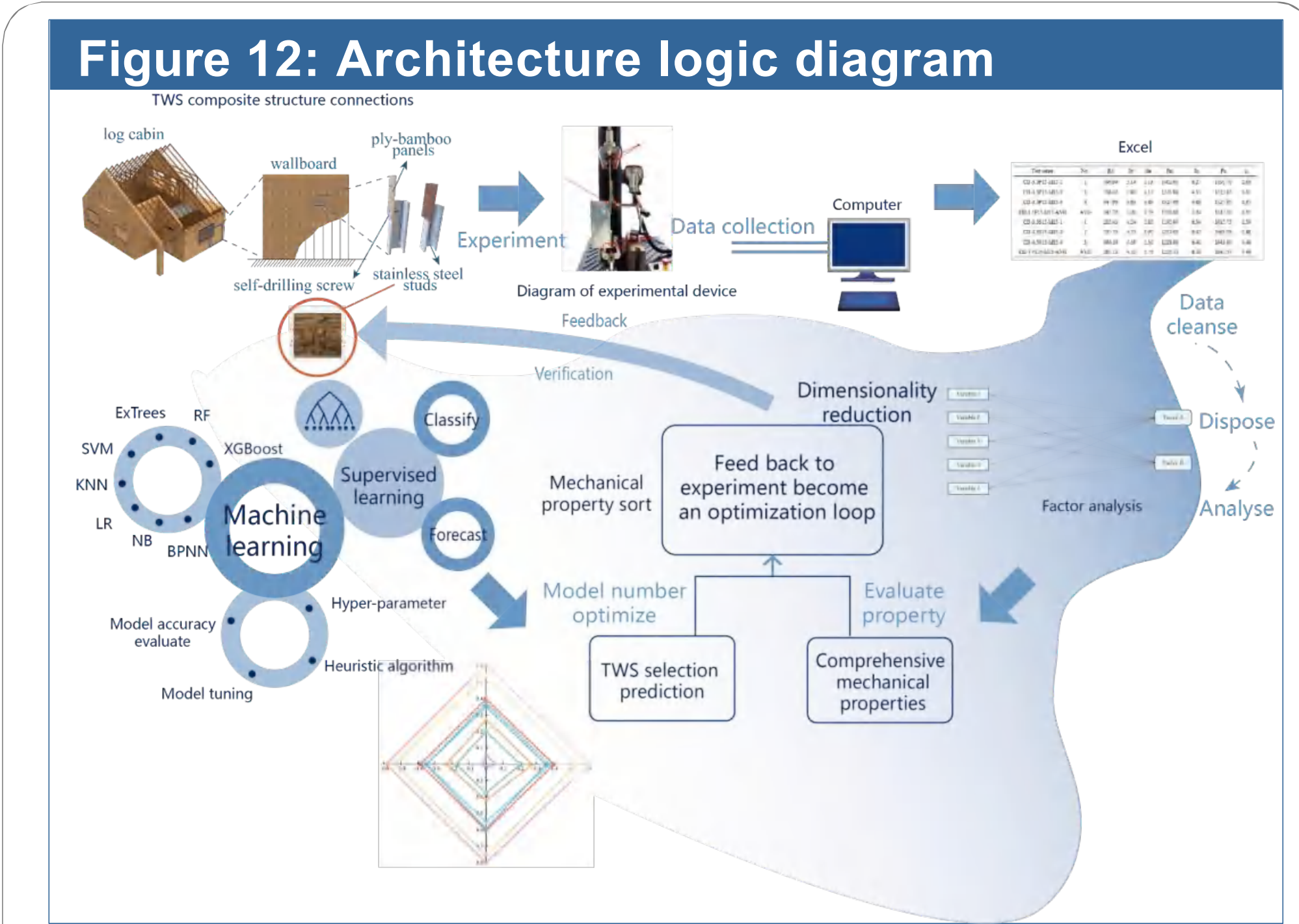
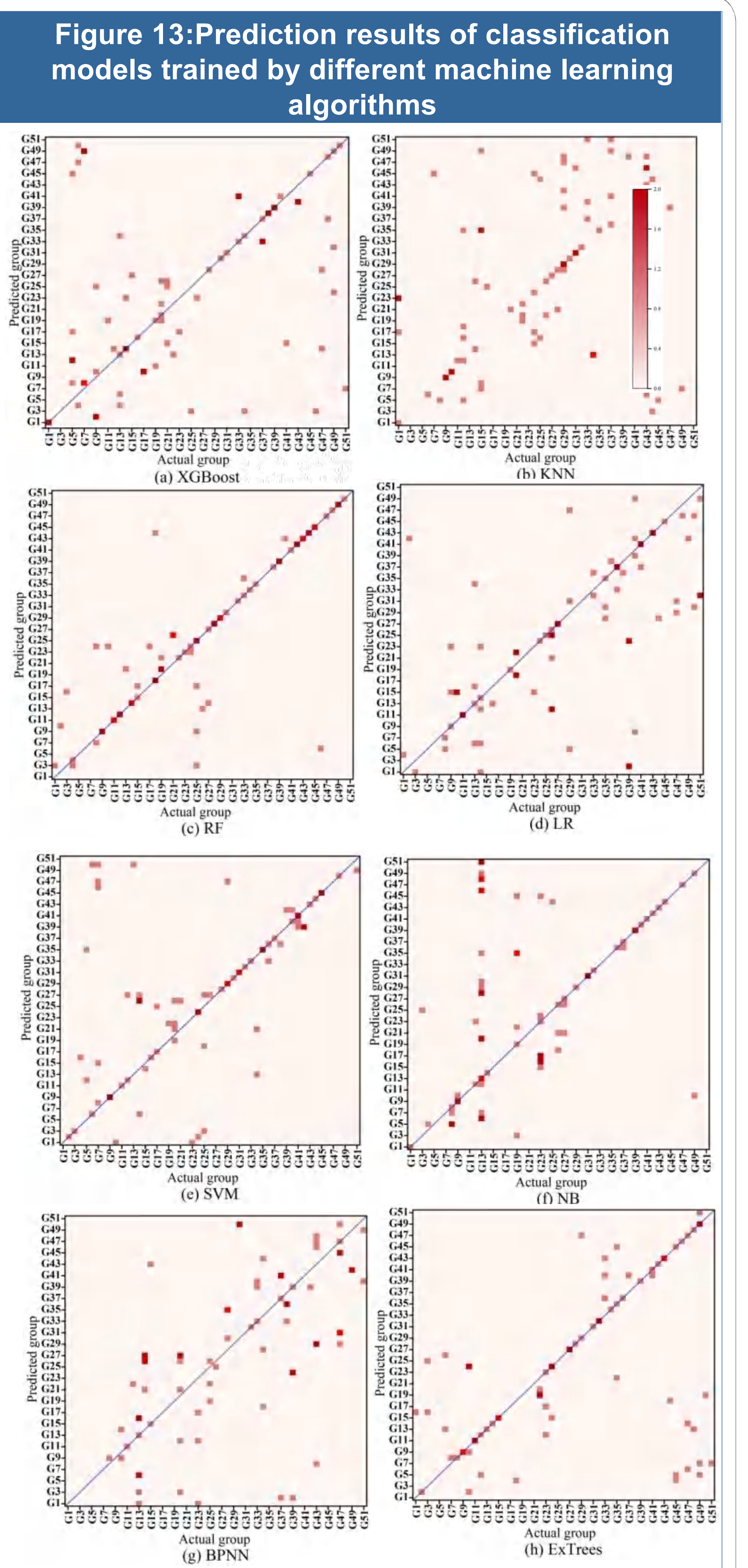
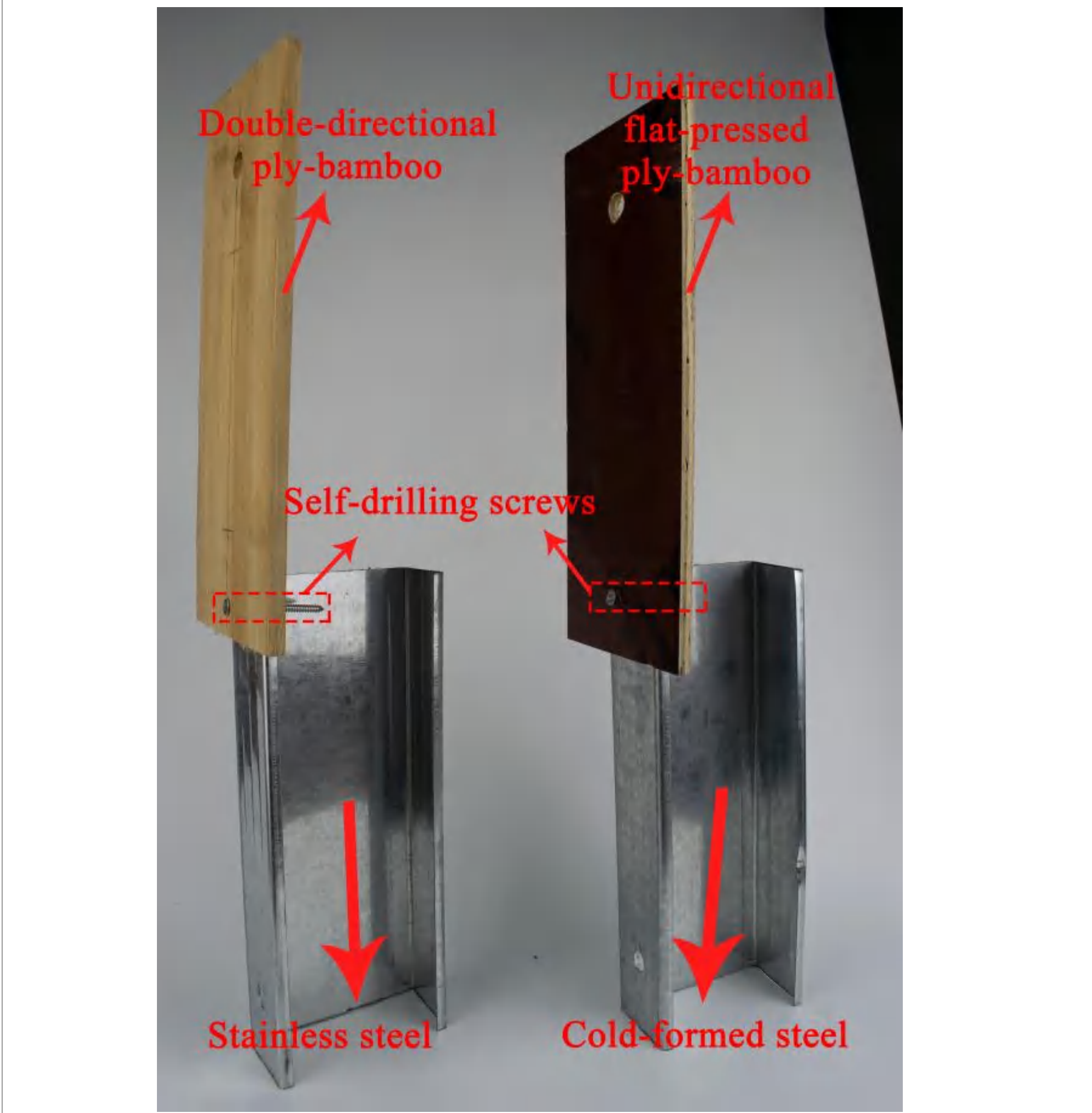
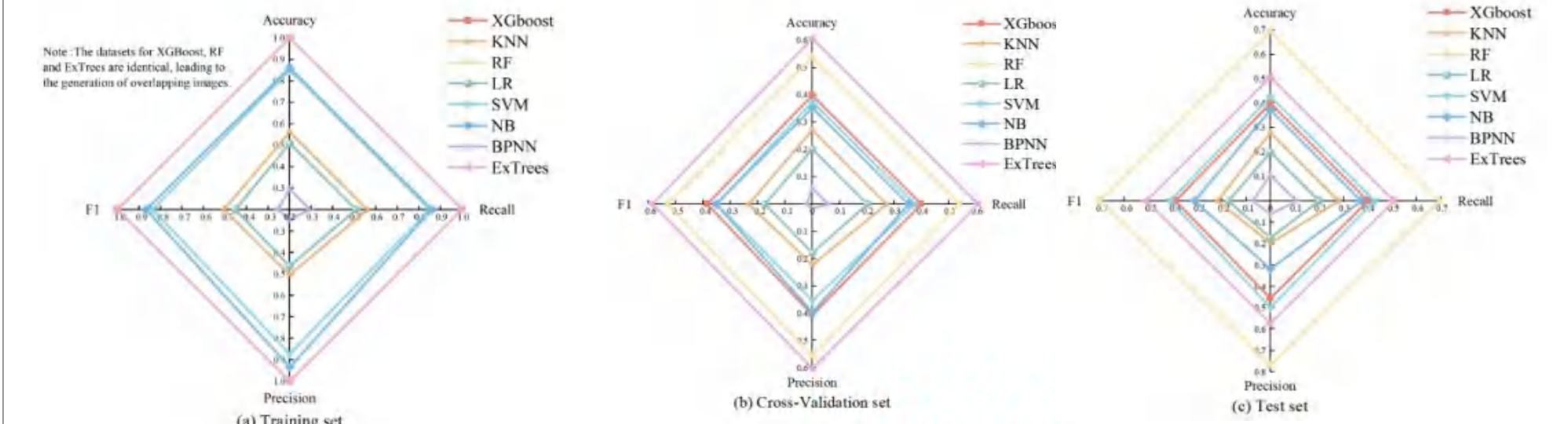


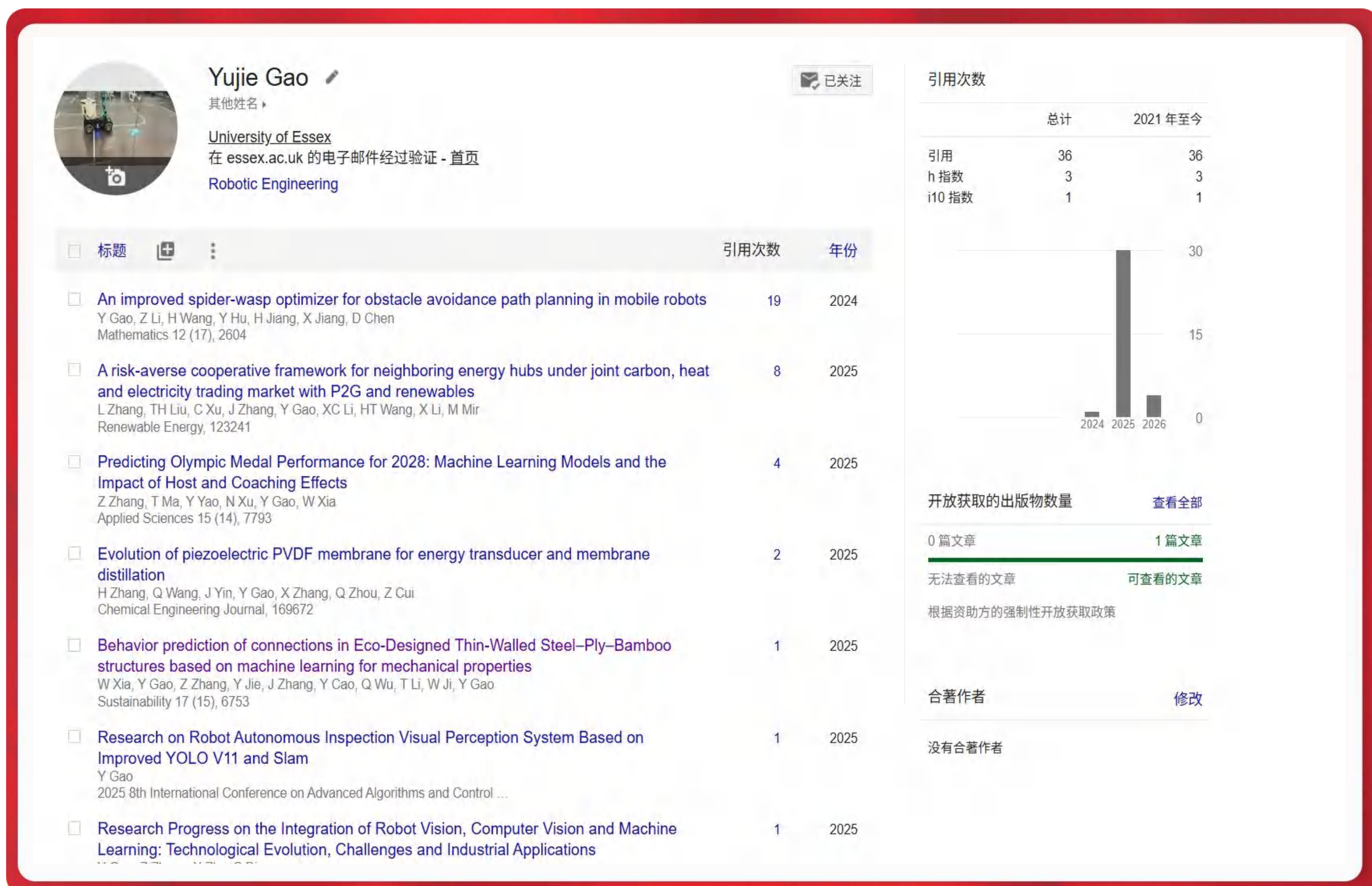
Table 3. Performance indicators of the eight different machine learning models

Model	Training				Cross-Validation				Test			
	AC	RE	PR	F1	AC	RE	PR	F1	AC	RE	PR	F1
XGBoost	1.000	1.000	1.000	1.000	0.399	0.399	0.406	0.385	0.400	0.400	0.456	0.390
KNN	0.578	0.578	0.542	0.528	0.272	0.272	0.223	0.233	0.280	0.280	0.197	0.216
RF	0.994	0.994	0.995	0.994	0.538	0.538	0.557	0.531	0.613	0.613	0.684	0.602
LR	0.439	0.439	0.344	0.363	0.202	0.202	0.179	0.174	0.200	0.200	0.172	0.172
SVM	0.821	0.821	0.839	0.814	0.376	0.376	0.353	0.351	0.427	0.427	0.498	0.411
NB	0.861	0.861	0.934	0.867	0.353	0.353	0.397	0.353	0.373	0.373	0.318	0.311
BPNN	0.295	0.295	0.245	0.255	0.058	0.058	0.016	0.021	0.107	0.107	0.061	0.072
ExTrees	0.994	0.994	0.995	0.994	0.608	0.608	0.602	0.586	0.507	0.507	0.573	0.509



• This paper proposed a new column theory framework based on machine learning (ML) technology. The shortcomings of structural type prediction for the matching performance of thin-walled steel bamboo structures were solved.
• A dataset of 249 measurement sets on the mechanical performance of TWS composite structure connections was compiled, utilizing 51 distinct types of connection specimens (using orthogonal experimental design and multiple machine learning methods to study the effects of tension, yield load, peak load, corresponding displacement, failure load, corresponding displacement and ductility coefficient of connections within TWS structures).
• Hyperparameter optimization

Sartup experience



- **Reviewer for Frontiers in Robotics and AI** (SCI journal|JCR Q2)
- **First author two sci papers (including top journals) and two ei conference papers**
- **Three founders of technology companies**, involved in fine chemicals, robotics, AI, and consumer-oriented smart products. IEEE Membership
- **Members of 9 interdisciplinary sustainability technology communities / associations including** IEEE Consultants Network Membership , IEEE Biometrics Council
- **Team leader of 20 national and provincial competitions.** Member of seven **volunteer teams**.
- **President / Manager of three national and university-level Entrepreneurship Associations.** Hosted 15+ charity entrepreneurship forums/speaking events, with a total of over 1,200 participants.
- **Five interdisciplinary integration projects**, one experience as a **Research Assistant** at Nanjing Tech University



- UI design,
- product delivery,
- signing orders,
- market research

- Search for **market gaps**, address **industry pain points**, and target customer needs

- AI+Data -Predictive **supply chain** modelingLightweight Tools

- Embodied Red Bird Challenge Camp philosophy through tech.enabled SME transformation



编号: DCY202408

入驻协议书

甲方: 江苏南大科技园有限公司

乙方: 南京中科聚联科技信息有限公司

甲方为南京市大学创新创业园(以下简称“示范园”), 位于南京市新模范马路 5 号的南京工业大学大东桥校区, 是南京工业大学与南京市人社局、鼓楼区政府本着“整合资源、优势互补、创新驱动、互动共赢”的原则, 共同建设的示范型园区。乙方申请在示范园建立企业的大学生团队。甲乙双方本着诚信、平等、自愿的原则, 经友好协商, 达成如下协议:

第一章 甲方的责任与义务:

1. 在示范园相关科技创新创业楼 B21-12 号工位, 为乙方提供办公场地, 使用期限为《2024 年 11 月 2 日至 2024 年 12 月 31 日》。租金为零。

2. 协助符合条件的企业申请政府科技经费等各类专项资金。(如取得大创企业可享受 3 年不多于 30 万创业租金减免补贴)

3. 协助企业办理各类证照, 落实工商和财税等优惠优惠政策。

4. 提供创业咨询和辅导、项目评估、商业计划完善等咨询服务。

5. 提供项目推荐申报、投融资信息、市场营销管理咨询服务。

6. 提供财务、税务、打印、传真、复印等行政事务外包服务。

7. 提供日常法律援助服务。

第二章 乙方的责任与义务:

1. 严格遵守示范园的相关规定, 爱护公物, 履行安全、卫生责任, 确保示范园的和谐有序。

2. 示范园系示范园, 履行企业主体责任。

3. 在生产经营活动中, 必须自觉遵守国家政策及法律法规, 入驻企业由甲方自主经营, 独立核算、自负盈亏。每年度提供企业财务报表(复印件), 包括资产负债表、损益表及现金流量附表等。每年度填写“企业年度经营报表”, 及时准确地向创业园报送不涉及经营机密的数据和报表, 配合园区相关统计工作。

4. 本协议签订后 15 个工作日内, 向甲方交纳房屋押金人民币 2000 元(大写: 贰仟元整)。

5. 不得擅自更改办公场地的水、电、网络、门窗及办公家具等设施设备; 办公场所有自然损坏需要维修的, 乙方应报备甲方, 由甲方派人维修; 若由于乙方使用不当或自行修缮造成房屋及家具、设施损坏的, 由乙方承担赔偿等所有责任。

6. 不得将办公场地转租、分租、转让、转借或擅自调换使用及与他人合租, 不得利用办公场地进行非法活动, 严禁进行宗教活动和传销。出现以上行为之一, 乙方在自己承担相应法律责任同时, 甲方有权终止协议并即收回办公场地, 乙方不承担任何费用。

7. 乙方不得占用公共区域、通道及指定空间, 不得擅自搭建、堆放物品。若有违反, 甲方有权终止协议, 并令其拆除、清理。

8. 乙方应在每季度的每一个季度的首月一次性缴纳当季度场地租金, 对房屋内所使用的水、电、电话、网络、物业管理、停车费和其他乙方消费的费用均由乙方支付。

9. 本协议生效后, 在确认乙方已签署并对外办公场地及家具设备非正常损坏并经甲方检查合格, 同时结清相关费用后, 乙方才能办理保证金退还等手续; 乙方撤离时, 应注意安全, 保护好甲方原有设施设备, 造成损坏, 乙方应承担赔偿责任。

10. 乙方擅自由本协议规定之日起 15 个工作日内入驻园区并开展业务活动。

第三章 甲方有权在下列情况出现后, 书面通知乙方终止或解除本协议, 乙方应通知如不纠正整改的, 甲方有权要求书面通知解除本协议并立即收回乙方承租或使用的办公场地, 并有权要求乙方承担违约责任:

1. 提供相关材料弄虚作假作假, 取消其资格, 并在三年内不得申请入驻;

2. 乙方不配合园区的工作安排, 产生违反园区的有关规章制度, 造成恶劣影响;

3. 乙方从协议签订之日起超过 15 个工作日内未入驻园区并开展业务;

4. 乙方未经甲方书面同意擅自将未入驻园区或创业团队搬离;

5. 未能按有关规定或合同约定按时交纳水、电、电话、网络、物业管理费等相关费用;

6. 乙方不能在协议签订后 3 个月内完成公司注册工商的;

7. 违反本协议约定的其他行为。

第四章 本协议有效期期限自甲方提供办公场所使用期限, 除以上第二条、第三条所列情况外, 任何一方要求提前终止本协议, 应提前 15 日告知。本协议有效期满前, 双方有续约的, 应在本协议有效期满前 30 日协商确定。

第五章 本协议有关规定的文字及印件与印制文字具有同等效力。本协议及规定的事项, 均遵守中华人民共和国有关法律、法规和规章制度。

本协议壹式叁份, 自签订之日起正式生效。

甲方(盖章): 乙方(盖章):

代表人: 代表人:

联系电话: 联系电话:

年 月 日 年 月 日

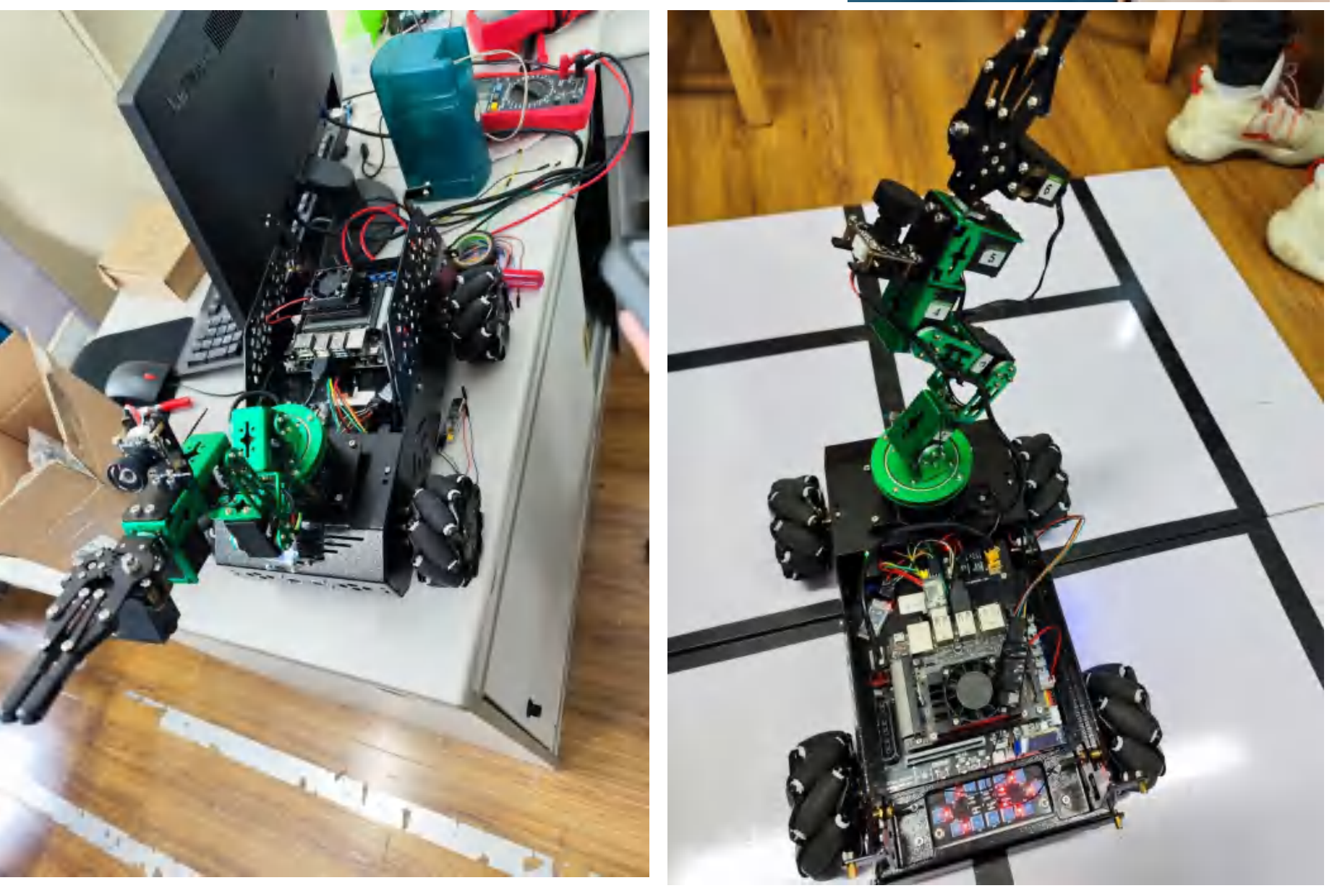
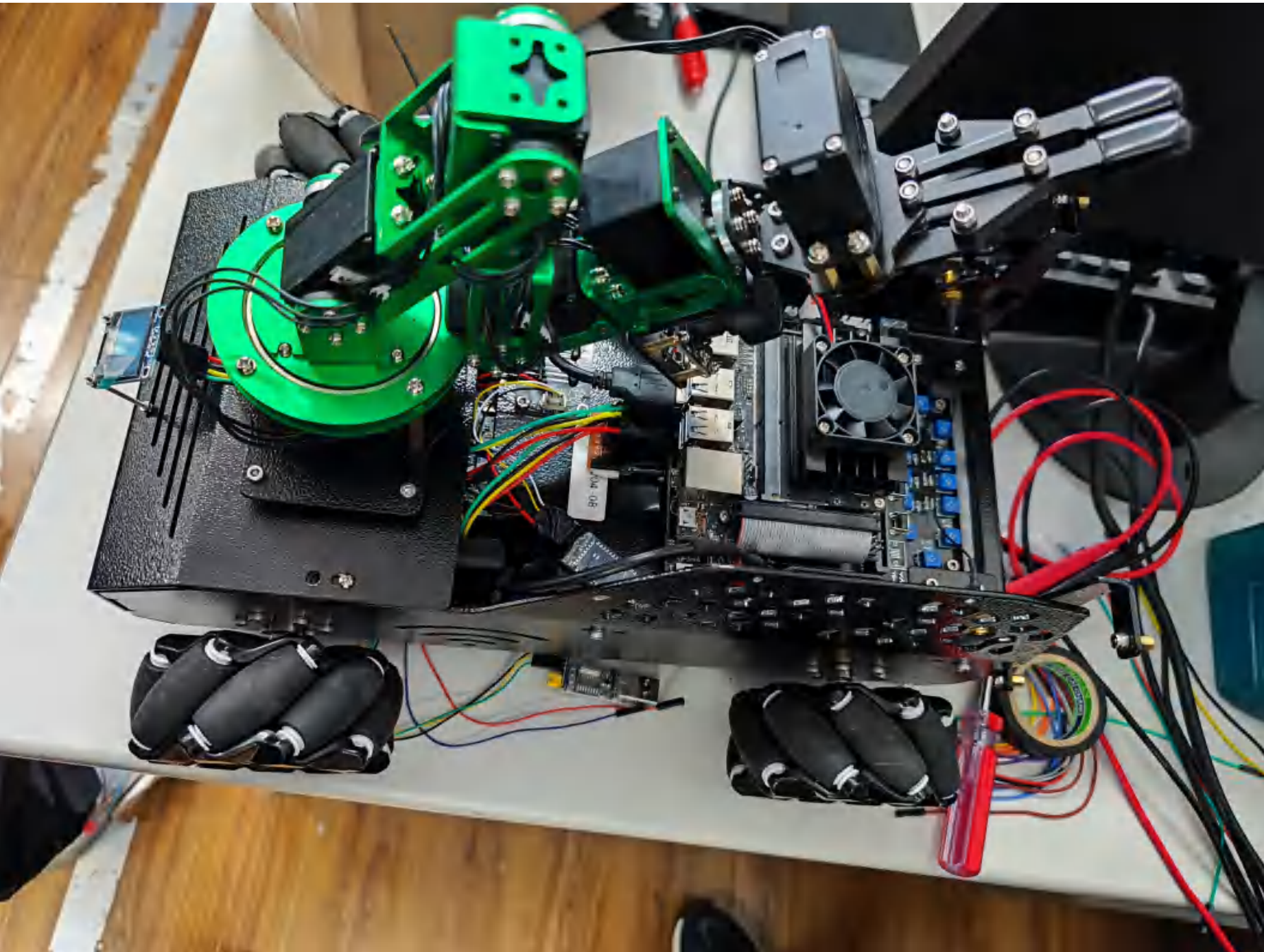
Contract for Entering the Industrial park

"Enable pet supplies to read minds"

Technical essence: Multimodal behavior recognition to understand animal needs

Technical Application Competition Awards and Technology Transformation Results

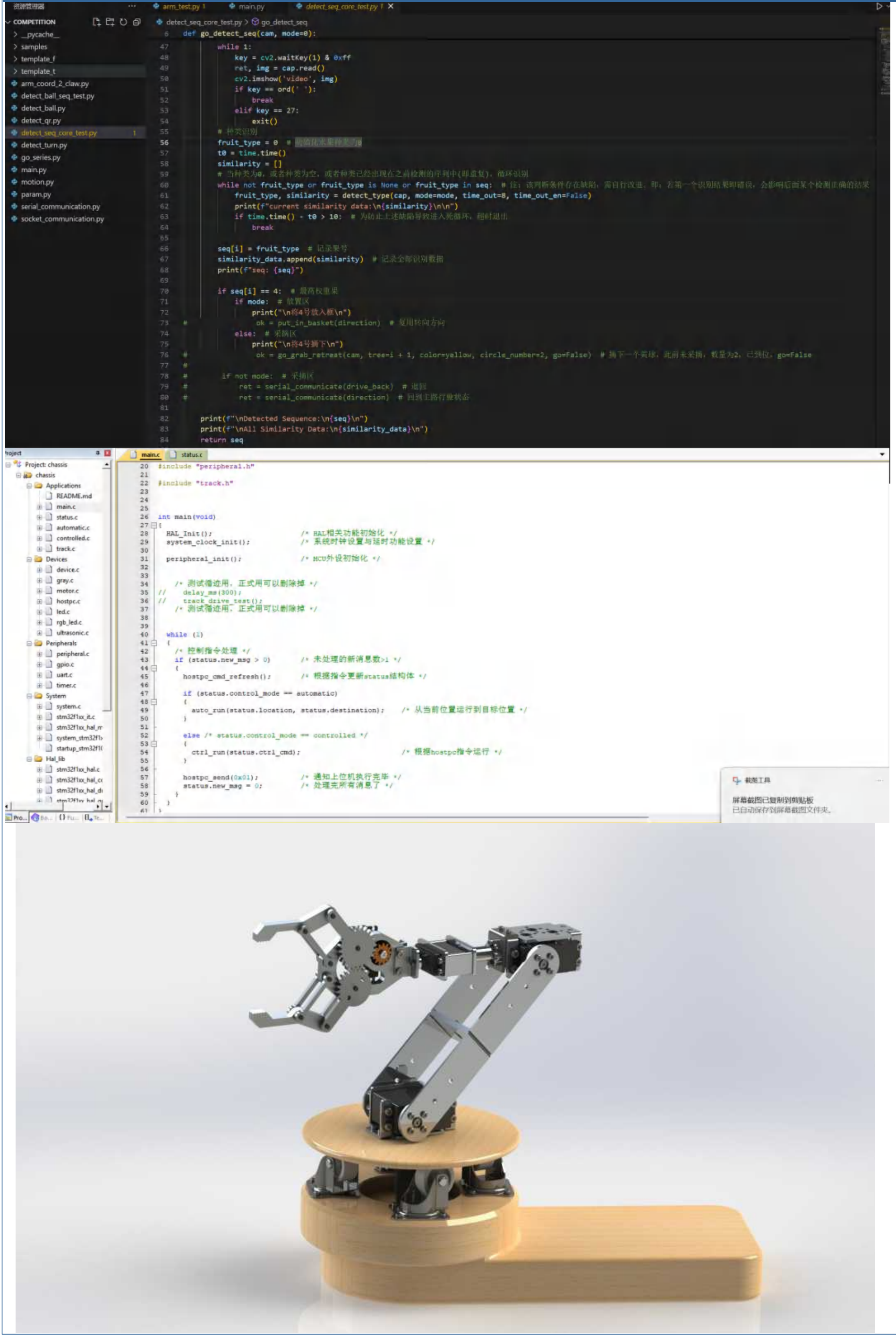
Figure 12: Smart Orchard Intelligent Fruit Picking Robot



RAIRCOM Developer Competition
AI + Agriculture + Robotics
National Champion
Smart Orchard Intelligent Fruit Picking Robot
• We developed an intelligent fruit-picking and autonomous navigation robot by integrating a dual-layer microcontroller architecture with NVIDIA Jetson Nano. The system combines machine vision

- and robotic control: the Jetson Nano handles real-time fruit recognition and environment mapping through edge computing, while the microcontrollers precisely coordinate the robotic arm's movement and vehicle navigation. This hybrid design enables seamless interaction between AI processing and electromechanical execution, achieving autonomous harvesting and path planning in orchard environments. The solution demonstrates scalable agricultural automation through hardware-software co-design, enhancing both productivity and operational safety. (Entire process involves 80% self-designed DIY.)

Figure 13: The design integrating software and hardware



- OpenCV recognizes fruit
- SW robotic arm mechanical design and 3D printing
- The lower machine chassis pid controls the robot arm and tire movement steering

Awards (SELECTED)



SCI paper title	Journal Name	SCI JCR Section s	Publish ed
An Improved Spider-Wasp Optimizer for Obstacle Avoidance Path Planning in Mobile Robots	Mathematic s	Q1	yes
Revealing internal β -phase PVDF membrane fouling tendency by double layer	Applied Water Science	Q1	In investm ent
A risk-averse cooperative framework for neighboring energy hubs under joint carbon, heat and electricity trading market with P2G and renewables	Renewable Energy	Q1	yes
Behaviour prediction of connections in thin-walled steel-ply-bamboo structures based on machine learning for mechanical properties	Sustainabil ity	Q1	Under review

Conference Publications

- Yujie Gao, Research on robot autonomous inspection visual perception system based on improved YOLO V11 and SLAM (ICAACE 2025 EI | IEEE | On-site oral report)
- Zhengkai Zhang, Yujie Gao, Dong Cheng, Exploring and Enhancing Advanced MoE Models: From Deepspeed-MoE to DeepSeek-V3(EI | IEEE AINIT 2025)
- Haoyue Ji, Yujie Gao, Enhancing Export Potential of Digital Service Trade in BRI Countries: A Stochastic Frontier Gravity Model Analysis of Digital Economy Development and Mediation Pathways(EI | ACM DEIC 2025)



Competition

- Captain of the National First Prize of RAICOM Robot Developer Competition
- The American College Student Mathematical Modeling (MCM/CM) h Award, the second prize in international competition, twice the captain
- China Robotics Innovation and Practice Forum National First Prize Captain Award
- College Students' Innovation and Entrepreneurship Project
- 15 provincial awards in finance, mathematical modeling, and robotics competitions
-etc



- Hard technology integration ability
- Complex problem abstraction ability
- Cross-border resource leadership ability

	patent name	First inventor	state
patent for invention	一种计算机电子器件加工装置	Gao Yujie	review
patent	一种用于智能识别的机器学习装置	Gao Yujie	accept
software copyright	LifeMate智能家居后台操作系统	Gao Yujie	accept
software copyright	人工智能一体化办公管控系统	Gao Yujie	accept

Short-term research plan

Time Period	Research Objectives	Publication Targets
2026.9 - 2027.2	- Complete the theoretical design of hierarchical vision transformers with dynamic pruning. - Conduct preliminary experiments on self-supervised representation learning with contrastive predictive coding. - Optimize the adaptive	
2027.3 - 2027.8	attention mechanisms with meta-learning in vision transformers.- Implement and test the unsupervised domain adaptation with GANs for robotic vision.	- Submit a journal paper to IEEE Transactions on Pattern Analysis and Machine Intelligence (SCI TOP) on the adaptive attention mechanisms in vision transformers.
2027.9 - 2028.2	- Refine the temporal modeling techniques with RNNsand memory-augmented networks. - Improve the self-supervised representation learning algorithms based on experimental results.	
2028.3 - 2028.8	- Develop the robust vision algorithms under uncertainty with Bayesian neural networks and Monte Carlo dropout. - Integrate all the developed algorithms into a comprehensive robotic vision system.	- Submit a journal paper to International Journal of Computer Vision (SCI TOP) on the robust vision algorithms for robotic vision.
2028.9 - 2029.2	- Conduct large-scale experiments on the integrated robotic vision system in different real-world scenarios. - Analyze the experimental results and make final improvements.	- Submit a conference paper to ECCV (CCF-A) on the performance evaluation of the integrated robotic vision system.
2029.3 - 2029.8	- Summarize the research findings and write the PhD thesis. - Prepare for the defense.	- Submit a journal paper to ACM Transactions on Graphics (SCI TOP) on the overall contributions of the research in robotic vision.

Academic industrialization



- Established a company and received recognition and investment from both the academic and business communities, establishing a submarine sensor constructed using a new type of pvdf membrane.

核心技术

南京中科智联科技有限公司
NANJING ZHONGKE ZHILIAN TECHNOLOGY CO., LTD.

合作开展β相PVDF薄膜成型相关工作 突破大规模极化工艺，实现压电膜可控连续化国产制备，成本下降35%

解决痛点一

实现偶极子反转

专利号

ZL 201710119413.5

涂膜方法

- 使用热流延等涂膜方法。膜厚度可从纳米级到毫米级，满足各个领域的要求
- 对薄膜产品进行电场极化加工，进一步提高压电元件的电介性能、机电耦合性等

专利号

202110440828.9

传统侧扫式声纳

有线通信

专利号

202421192983.9

202310824179.1

202421193155.7

第一负责人

高宇杰

高宇杰

高宇杰

新型潜艇探测器

自由下潜深度增加十倍！

水声通信

多传感器感知模式

重量减小44%

解决痛点二：下潜更深 更高质量的扫描图像

供应体系

技术指导

原材料供应链

主体业务

中科智联

目标客户

数据服务

整机销售

核心部件 自研自产

致力于成为海洋环境勘探领域的“华为”，解决深海勘探问题

公司成立并
稳定运行

单个技术订单银行
实际到账20w

南京中科智联科技有限公司成立并获得信用认证

结成战略合作伙伴

客户反馈良好

验资报告

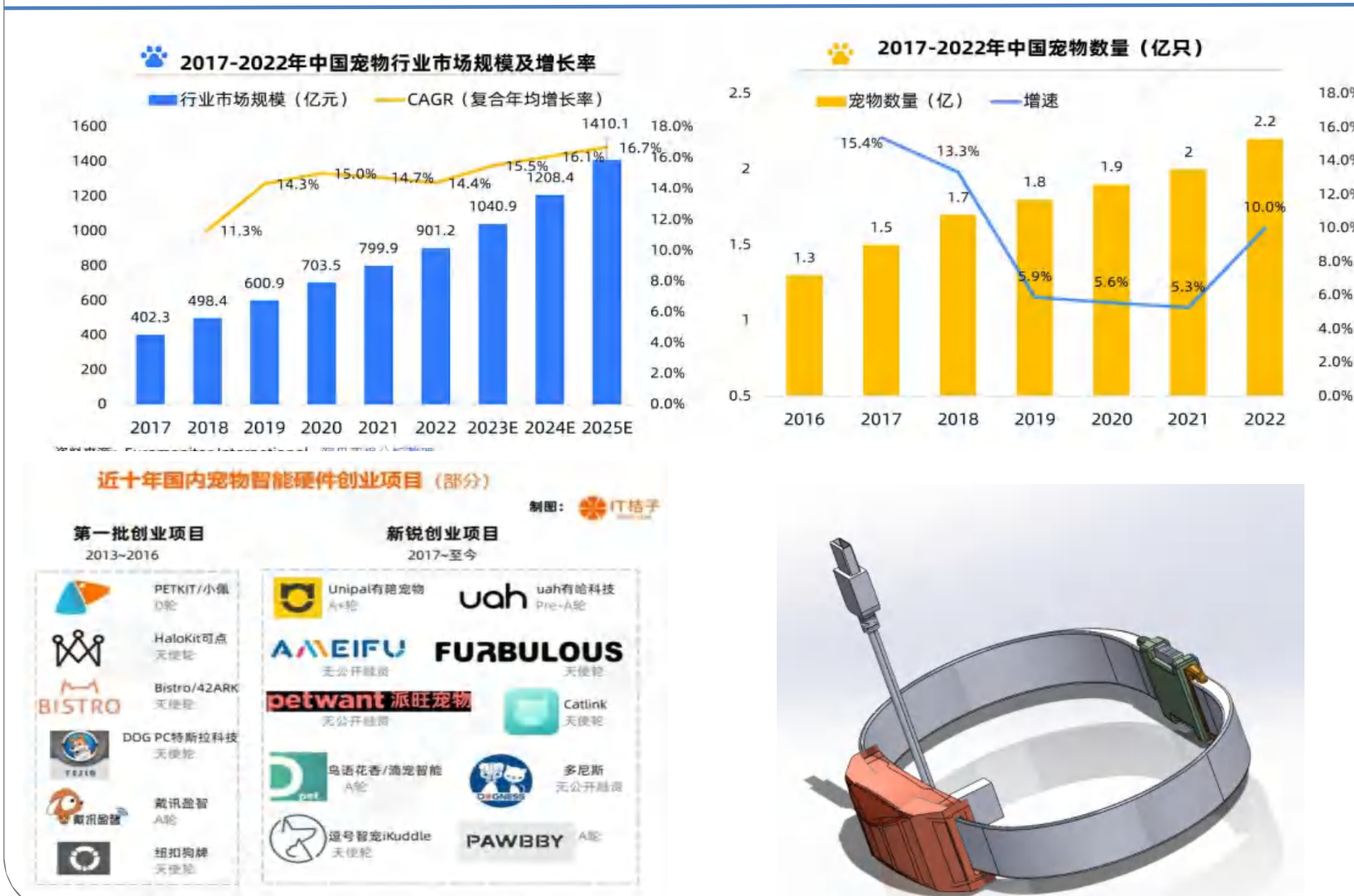
技术服务案例

- Cooperative unit:
- **National Engineering Research Center of Membrane Technology**
 - **Shanghai Ocean University**

- Technical development stage

- The establishment of the company
- The determination of shares
- The credit and qualification of the enterprise

- The establishment of the company
- The determination of shares
- The credit and qualification of the enterprise



Volunteer / Serve the Society

南京圆梦志愿服务中心函件

理事委员会关于邀请社会公益及青年学生界别委员的函

根据团中央材料，团区委金秘书长及团区委书记，邀请您担任理事委员会委员，担任南京圆梦志愿服务中心青年学生界别、增选监事。

此致问候，感谢贵单位，并快会贵单位关系，如不中意的贵单位会告知。

南京圆梦志愿服务中心

高宇杰

4

4

4

4

27

4

118

999+

参加项目

参加团体

服务时长

个人排名

志愿服务证

奉献、友爱、互助、进步

姓名 高宇杰

性别 男

南京圆梦志愿服务中心

南京圆梦志愿服务中心

服务区域

城南新区

注册日期

2021-04-14

南京圆梦志愿者服务中心理事委员会

南京市江北新区圆梦社会工作服务中心

2023年度南京圆梦志愿者服务中心

理事委员会委员入选程序

一、委员工作方式

高宇杰

A型 (Rh+)

点击刷新后已档案

3次

1000 ml

全血

0次

成分血 0 治疗量

最近

已是最新记录

有疑问?

26日

捐献 全血 300 ml

04月

◎ 南京 · 献血单位 第3次

>

👍 0

2023年

19日

捐献 全血 400 ml

10月

◎ 南京 · 献血单位 第2次

>

👍 0

2022年

20日

捐献 全血 300 ml

10月

◎ 南京 · 献血单位 第1次

>

👍 0

2023

红蚂蚁圆梦慈善基金会(南京总部)

南京圆梦社会工作服务中心

2023.9-2023.12

申请 · 报名 · 面试

2023

红蚂蚁圆梦慈善基金会(南京总部)

南京圆梦社会工作服务中心

2023.9-2023.12

申请 · 报名 · 面试

2023

红蚂蚁圆梦慈善基金会(南京总部)

南京圆梦社会工作服务中心

2023.9-2023.12

申请 · 报名 · 面试

- Every year, at a fixed time, I **donate blood voluntarily** to contribute to society.

[illegible]

- program at the Shenzhen Institute of Automation of the Chinese Academy of Sciences. **(Embodied AI)**

- **President of Association** since June 01, 2024
Technology elite entrepreneurs Association, Nanjing Tech University
- **Member of the pre-creation Lab**
November 01, 2024– November 01, 2024
Nanjing Tech University National college students Pioneer
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The School-Enterprise Integration Professional Committee of the
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- **Academic service / Reviewer** since March 01, 2025
Frontiers

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 ORCID: <https://orcid.org/0009-0003-2071-4262>

Google Scholar: <https://scholar.google.com.hk/citations?hl=zh-CN&user=E5rvKWgAAAAJ>
WOS: <https://webofscience.clarivate.cn/wos/author/record/NCV-7572-2025> (updating)

Thanks for watching and correcting